

Comprovante do artigo Biometrical Journal 2019

Categoria: Artigos

Regra: 2.9 - 15 pontos por publicação

Pontos lançados: 15.00

Received: 12 March 2018 | Revised: 17 October 2018 | Accepted: 22 February 2019
 DOI: 10.1002/bimj.201800056



Biometrical Journal

RESEARCH PAPER

Defective regression models for cure rate modeling with interval-censored data

Item 2.9

Vinicius F. Calsavara¹ | Agatha S. Rodrigues^{2,3} | Ricardo Rocha⁴ |
 Vera Tomazella⁵ | Francisco Louzada⁶

¹Department of Epidemiology and Statistics, A.C. Camargo Cancer Center, São Paulo, SP, Brazil

²Institute of Mathematics and Statistics, University of São Paulo, São Paulo, SP, Brazil

³Department of Obstetrics and Gynecology, São Paulo University Medical School, São Paulo, SP, Brazil

⁴Department of Statistics, Federal University of Bahia, Salvador, BA, Brazil

⁵Department of Statistics, Federal University of São Carlos, São Carlos, SP, Brazil

⁶Institute of Mathematical Science and Computing, University of São Paulo, São Carlos, SP, Brazil

Correspondence

Vinicius F. Calsavara, Department of Epidemiology and Statistics, A.C. Camargo Cancer Center, São Paulo, SP, Brazil.
 Email: vinicius.calsavara@accamargo.org.br

Abstract

Regression models in survival analysis are most commonly applied for right-censored survival data. In some situations, the time to the event is not exactly observed, although it is known that the event occurred between two observed times. In practice, the moment of observation is frequently taken as the event occurrence time, and the interval-censored mechanism is ignored. We present a cure rate defective model for interval-censored event-time data. The defective distribution is characterized by a density function whose integration assumes a value less than one when the parameter domain differs from the usual domain. We use the Gompertz and inverse Gaussian defective distributions to model data containing cured elements and estimate parameters using the maximum likelihood estimation procedure. We evaluate the performance of the proposed models using Monte Carlo simulation studies. Practical relevance of the models is illustrated by applying datasets on ovarian cancer recurrence and oral lesions in children after liver transplantation, both of which were derived from studies performed at A.C. Camargo Cancer Center in São Paulo, Brazil.

KEYWORDS

defective distribution, Gompertz distribution, inverse Gaussian distribution, interval-censored data, long-term survivor

1 | INTRODUCTION

In survival data, it is often of interest to model the time T until an event (e.g., cancer recurrence) occurs; but in some situations, time T may only be known to have occurred within an interval $(U, V]$, where $U < T \leq V$. For instance, a liver transplant patient may develop an oral lesion that is detected during a follow-up visit after surgery. In this situation, patients submitted to liver transplant are assessed at clinic visits by a dentist in order to detect oral lesion. The time to the event (oral lesion) is not observed, but the event is known to have happened between time U and time V .

In practice, researchers typically perform traditional survival analysis methods using the date when the event is diagnosed (diagnostic exam) as the moment of event occurrence. However, if an event occurs before time V but after time U , ignoring the interval-censored mechanism in survival analysis can lead to overestimates in survival curves and erroneous conclusions (Rodrigues, Calsavara, Silva, Alves, & Vivas, 2018).

Researchers have proposed some approaches for handling such interval-censored data. Peto (1973) provided an estimation method for the cumulative distribution function, which is similar to the life-table technique, as well as the self-consistency algorithm for the estimation of the survival function proposed by Turnbull (1976). Semiparametric approaches based on the